

average profile of publication placement along the journal hierarchy, regardless of the specificity of individual disciplines. The shape of the distributions alone reveals a strong quality gradient.

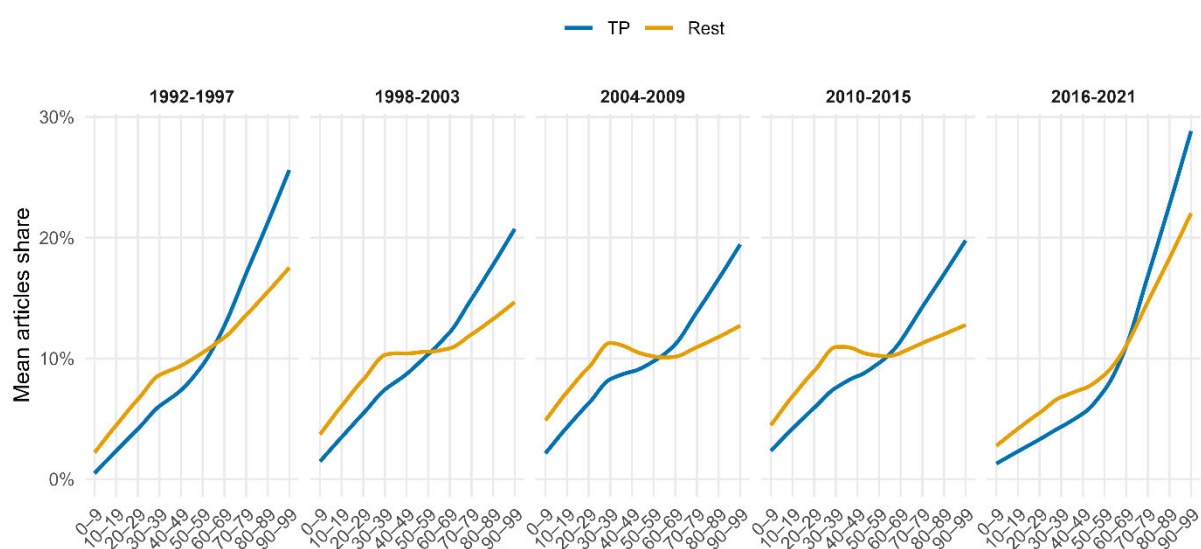
In each period analyzed, for both top performers and other scientists, the share of articles systematically increases with the transition to higher journal deciles. This means that, on average, publications are more often placed in stronger publishing channels (which reflects the general orientation of the Polish science system toward more prestigious journals).

In the lower and middle segments of the distribution (up to about the 50–60 percentiles), the profiles of both groups are similar. The curves of the two classes run almost parallel (and in some places overlap). This suggests that publishing in low- and medium-quality journals is not the most important measure of stratification between top performers and other authors.

A clear divide only appears at the top of the journal hierarchy. In the three upper percentiles (70–79, 80–89, and especially 90–99), the top performer curve consistently separates from the curve for the rest. This difference is observable in every period and gradually increases over time. The strongest contrast is visible in the years 2016–2021, where the share of top performers' articles in the highest decile exceeds the share of the other group by several, and in some places even more than a dozen, percentage points. At the same time, it can be seen that the concentration of publications in top journals is growing for both groups, but the pace of this growth is clearly faster among top performers.

Overall, it is clear that the differences between top performers and other authors are not evenly distributed across the entire publication spectrum. They are evident almost exclusively at the top of the journal hierarchy, while in the lower and middle deciles, the distributions of both groups remain very similar. As a result, stratification in science is primarily qualitative in nature. It is driven by selective access to the most prestigious publication channels.

Figure 6. Article average percentages by decile (all disciplines combined)



4.2. Model approach

The model is estimated using a micro-panel approach, where the unit of observation is a scientist active in a specific period and discipline (scientist $\times$ period $\times$ discipline). This data structure

allowed us to track changes in scientists' positions over time and analyze them within comparable disciplinary contexts (see Table 6).

The dependent variable is a binary indicator of membership in the top performer class (the upper 10% in terms of journal prestige-normalized productivity). It takes the value 1 when the scientists is in the top 10% of the quality-weighted productivity distribution in a given reference cell (period $\times$ discipline), and 0 in all other cases. In other words, the model does not explain the level of productivity itself but rather the probability of entering the publishing elite under specific temporal and disciplinary conditions.

Of key importance is the construction of the journal prestige-normalized productivity measure, which forms the basis for determining top performer status by rewarding publications in the most prestigious journals and giving them disproportionately greater weight in the scientists' ranking.

The model was estimated as a generalized linear model with mixed effects (GLMM), with a function that combined a logistic function and a binomial distribution. Such a specification is natural for a binary dependent variable and allows for direct modeling of the probability of being in the top performer class, while taking into account the panel structure of the data.

The fixed part of the model was designed around two basic dimensions of scientific activity: the quantity and quality of research production. The quantitative dimension is represented by the logarithm of the number of publications by an author in a given cell ($\ln_pubs_c$) and its quadratic term. This allowed us to capture the potential nonlinearity of the relationship. The qualitative dimension describes the mean percentile of journals in which the author published ($mean_journal_pct_c$). A key element of the specification is the interaction between quantity and quality ($\ln_pubs_c:mean_journal_pct_c$), which permitted us to directly answer the question of whether high publishing intensity in average publication channels is sufficient to enter the elite of top performers, or whether locating articles in the most prestigious journals is of decisive importance.

The model also takes into account a set of demographic and institutional variables, such as gender (Male/Male), academic age ($acage_c$), institution type ($inst_typeRest$), average team size calculated on the basis of all co-authorships in a period (ATS_c), and a median prestige of publication channels, also calculated separately for each period ($prestmed_c$). Gender and academic age variables appear both as main effects and as moderators of the relationship between quantity and quality and the probability of entering the TOP class, which allows for testing differences in the mechanisms of selection to the publishing elite between the two classes of authors. We calculated academic age as the time passed from the first publication in the Scopus dataset (any type) to the last year of each study period. The description of independent and fixed effects variables is shown in Table 4, and the variables included in the model are presented in Table 5.

Table 4. Independent and fixed effects variables

Variable (model term)	Readable name	Type/Coding
(Intercept)	Baseline log-odds	Constant
Period 1998–2003	Period: 1998–2003	Dummy (ref. 1992–1997)
Period 2004–2009	Period: 2004–2009	Dummy (ref. 1992–1997)
Period 2010–2015	Period: 2010–2015	Dummy (ref. 1992–1997)
Period 2016–2021	Period: 2016–2021	Dummy (ref. 1992–1997)
Discipline: BIO	Biology	Dummy (ref. AGRI)

Variable (model term)	Readable name	Type/Coding
Discipline: CHEM	Chemistry	Dummy (ref. AGRI)
Discipline: CHEMENG	Chemical Engineering	Dummy (ref. AGRI)
Discipline: COMP	Computer Science	Dummy (ref. AGRI)
Discipline: EARTH	Earth Sciences	Dummy (ref. AGRI)
Discipline: ENER	Energy	Dummy (ref. AGRI)
Discipline: ENG	Engineering	Dummy (ref. AGRI)
Discipline: ENVI	Environmental Sciences	Dummy (ref. AGRI)
Discipline: MATER	Materials Science	Dummy (ref. AGRI)
Discipline: MATH	Mathematics	Dummy (ref. AGRI)
Discipline: MED	Medicine	Dummy (ref. AGRI)
Discipline: NEURO	Neuroscience	Dummy (ref. AGRI)
Discipline: PHARM	Pharmacology	Dummy (ref. AGRI)
Discipline: PHYS	Physics	Dummy (ref. AGRI)
ln_pubs_c	Log number of publications	Continuous (centered)
ln_pubs2_c	Squared log publications	Continuous (centered)
mean_journal_pct_c	Mean journal percentile	Continuous (centered)
coop_c	Collaboration intensity	Continuous (centered)
coopint_c	International collaboration	Continuous (centered)
MaleMale	Male	Dummy (1 = male)
acage_c	Academic age	Continuous (centered)
inst_typeRest	Non-research-intensive institution	Dummy (ref. IDUB)
ATS_c	Academic time structure	Continuous (centered)
prestmed_c	Institutional prestige (median)	Continuous (centered)
ln_pubs_c:mean_journal_pct_c	Quantity $\times$ quality interaction	Interaction

Table 5. Variables in the model

Variable (model term)	Description
(Intercept)	Log-odds of being a top performer in the reference period (1992–1997), reference discipline (AGRI), and at mean values of all centered continuous variables.
Period 1998–2003	Publication period indicator capturing cohort and structural time effects relative to 1992–1997.
Period 2004–2009	Publication period indicator capturing cohort and structural time effects relative to 1992–1997.
Period 2010–2015	Publication period indicator capturing cohort and structural time effects relative to 1992–1997.
Period 2016–2021	Publication period indicator capturing cohort and structural time effects relative to 1992–1997.
Discipline: BIO	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: CHEM	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: CHEMENG	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: COMP	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: EARTH	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: ENER	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: ENG	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: ENVI	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: MATER	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: MATH	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: MED	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: NEURO	Discipline fixed effect capturing structural differences in publication regimes.
Discipline: PHARM	Discipline fixed effect capturing structural differences in publication regimes.

Variable (model term)	Description
Discipline: PHYS	Discipline fixed effect capturing structural differences in publication regimes.
ln_pubs_c	Natural logarithm of the total number of publications authored in a given period, mean-centered.
ln_pubs2_c	Squared term of the centered log number of publications, allowing for nonlinear scale effects.
mean_journal_pct_c	Average percentile rank of journals in which an author publishes, capturing publication quality.
coop_c	Measure of co-authorship intensity, capturing the extent of collaborative publishing.
coopint_c	Share or intensity of international co-authorships.
MaleMale	Gender indicator, with female as the reference category.
acage_c	Years since first publication (academic career age), mean-centered.
inst_typeRest	Indicator for employment outside the group of research-intensive (IDUB) universities.
ATS_c	Measure of academic employment intensity or workload allocation.
prestmed_c	Median prestige of the author's institutional affiliation.
ln_pubs_c:mean_journal_pct_c	Interaction between publication volume and journal quality, testing whether quantity effects depend on quality.

All quantitative variables were centered (around the mean), which facilitated the interpretation of main effects in the presence of interactions and reduced collinearity between linear and nonlinear components. In particular, it allowed main effects to refer to a scientist with an average level of other characteristics.

Structural heterogeneity is controlled by including fixed effects for period and discipline so that scientists are compared against specific historical and disciplinary conditions. At the same time, the random structure of the model includes a random author effect (1 | Auth_ID). This allows for control of unobserved, individual heterogeneity (such as permanent differences in abilities, research preferences, or network position).

Table 6. Statistics for the TOP 10% model

term	Estimate	Std.Error	z.value	p.value
(Intercept)	-6.654	0.149	-44.653	0.000
Period 1998–2003	-3.154	0.095	-33.165	0.000
Period 2004–2009	-5.909	0.109	-54.442	0.000
Period 2010–2015	-9.781	0.139	-70.238	0.000
Period 2016–2021	-16.161	0.202	-80.097	0.000
Discipline: BIO	-2.817	0.101	-27.927	0.000
Discipline: CHEM	-7.221	0.129	-55.952	0.000
Discipline: CHEMENG	6.654	0.224	29.640	0.000
Discipline: COMP	7.051	0.182	38.760	0.000
Discipline: EARTH	3.774	0.132	28.543	0.000
Discipline: ENER	8.071	0.256	31.472	0.000
Discipline: ENG	6.122	0.123	49.741	0.000
Discipline: ENVI	3.845	0.136	28.374	0.000
Discipline: MATER	-2.607	0.124	-21.027	0.000
Discipline: MATH	2.221	0.134	16.607	0.000
Discipline: MED	-1.227	0.080	-15.406	0.000
Discipline: NEURO	-2.010	0.224	-8.987	0.000
Discipline: PHARM	-1.854	0.275	-6.736	0.000
Discipline: PHYS	-8.574	0.144	-59.709	0.000

ln_pubs_c	10.843	0.238	45.601	0.000
ln_pubs2_c	3.762	0.119	31.585	0.000
mean_journal_pct_c	0.064	0.005	12.786	0.000
coop_c	0.517	0.159	3.254	0.001
coopint_c	0.087	0.077	1.123	0.261
MaleMale	0.012	0.046	0.266	0.791
acage_c	0.010	0.002	4.373	0.000
inst_typeRest	-0.137	0.047	-2.896	0.004
ATS_c	-0.008	0.005	-1.789	0.074
prestmed_c	0.017	0.003	5.665	0.000
ln_pubs_c:mean_journal_pct_c	0.185	0.004	42.086	0.000

The estimated logistic model was designed to determine what quantitative, qualitative, and structural mechanisms shape the probability of belonging to the group of top performers in science, and whether the quantity and quality of publication output act as substitute factors or instead reinforce each other.

The binary nature of the dependent variable, which identifies entry into the upper segment of the productivity distribution, justifies the use of a logit-type link function. All continuous variables were centered, and the number of publications was logarithmically transformed and expanded by a quadratic component. This facilitated the interpretation of parameters, limited the influence of extreme observations, and allowed for the capture of potential nonlinearities in the relationship.

The free term of the model takes the value -6.654 ($SE = 0.149$; $z = -44.653$; $p < 0.001$) and refers to a hypothetical scientist belonging to all reference categories of qualitative variables and characterized by average values of continuous variables. The high absolute negative value of this parameter indicates that the baseline probability of being in the top performer class is very low, which well reflects the publishing elite and the relative nature of the category.

The effects of time periods reveal a strong and systematic decline in the chances of achieving top performer status in subsequent time intervals relative to the reference period (1992–1997). In 1998–2003, the log-odds ratio is lower by 3.154 ($SE = 0.095$; $z = -33.165$; $p < 0.001$), in 2004–2009 by 5.909 ($SE = 0.109$; $z = -54.442$; $p < 0.001$), between 2010 and 2015 by 9.781 ($SE = 0.139$; $z = -70.238$; $p < 0.001$), and in 2016–2021 by as much as 16.161 ($SE = 0.202$; $z = -80.097$; $p < 0.001$). The monotonous increase in the absolute magnitude of these effects indicates that, with individual and structural characteristics remaining constant, entry into the publishing elite becomes increasingly difficult over time. This result suggests growing competition and relativity in selection for the publishing elite and provides a useful background for the interpretation of quantitative and qualitative effects.

The model also reveals striking cross-disciplinary differences. Compared to the reference field (AGRI), significantly lower chances of achieving top performer status are observed in, e.g., PHYS ($\beta = -8.574$; $SE = 0.144$; $z = -59.709$; $p < 0.001$), CHEM ($\beta = -7.221$; $SE = 0.129$; $z = -55.952$; $p < 0.001$), and BIO ($\beta = -2.817$; $SE = 0.101$; $z = -27.927$; $p < 0.001$). In turn, significantly higher odds of belonging to the publishing elite are observed in ENER, COMP, CHEMENG, and ENG, as well as in EARTH, ENVI, and MATH. The scale of these differences confirms that disciplinary context constitutes a permanent structural constraint (or support) for entry into the publishing elite, regardless of the individual scale and quality of production.

The central element of the model, directly related to our research questions, is the impact of quantitative productivity, journal quality, and their mutual interaction. The logarithm of the number